



SINGLE ANSWER TYPE

- $Na_2B_4O_7 \cdot 10H_2O \xrightarrow{\text{Heat}} X + NaBO_2 + H_2O$, $X + Cr_2O_3 \xrightarrow{\text{Heat}} Y$ (Green coloured) X and Y are:
A) Na_3BO_3 and $Cr(BO_2)_3$ B) $Na_2B_4O_7$ and $Cr(BO_2)_3$ C) B_2O_3 and $Cr(BO_2)_3$ D) B_2O_3 and $CrBO_3$
- Borax in its crystal possess:
A) 3 tetrahedral unit B) 2 tetrahedral and 2 planar triangular units
C) 3 tetrahedral and 2 planar triangular units D) all tetrahedral units
- Borax is converted into crystalline boron by the following steps:
$$\text{Borax} \xrightarrow{X} H_3BO_3 \xrightarrow{\Delta} B_2O_3 \xrightarrow[Y]{\Delta} B$$

 X and Y are respectively:
A) HCl , Mg B) HCl , C C) C , Al D) HCl , Al
- The dissolution of $Al(OH)_3$ by a solution of $NaOH$ results in the formation of:
A) $[Al(H_2O)_4(OH)_2]^+$ B) $[Al(H_2O)_3(OH)_3]$ C) $[Al(H_2O)_2(OH)_4]^-$ D) $[Al(H_2O)_6(OH)_3]$
- Anhydrous aluminium chloride fumes in moist air owing to the formation of:
A) Gaseous aluminium chloride B) Chlorine
C) Chlorine dioxide D) Hydrogen chloride
- Which species does not exist ?
A) $[BF_6]^{3-}$ B) $[AlF_6]^{3-}$ C) $[GaF_6]^{3-}$ D) $[InF_6]^{3-}$
- Which of the following cannot liberate H_2 with acids ?
A) Al B) In C) Ti D) B
- Which of the following is not an ionic trihalide ?
A) AlF_3 B) BF_3 C) InF_3 D) GaF_3
- When orthoboric acid (H_3BO_3) is heated strongly the residue left is :
A) boron B) metaboric acid C) boric anhydride D) borax
- Borax when heated with Pt wire forms a glass bead which is made up of
A) sodium metaborate B) sodium tetraborate
C) sodium metaborate and boric anhydride D) sodium tetraborate and boric anhydride
- The reaction, $B(OH)_3 + NaOH \rightarrow Na[B(OH)_4]$ can be made to proceed in forward direction by :
A) adding *cis*-1,2-diol B) adding borax C) adding *trans*-1,2-diol D) adding Na_2HPO_4
- The hybridization of boron atom in orthoboric acid is:
A) sp B) sp^2 C) sp^3 D) sp^3d
- Which is false in case of boric acid (H_3BO_3) ?
A) It is soluble in hot water B) It acts as a tribasic acid
C) It has a planar structure D) It acts as a monobasic acid
- The main factor responsible for weak acidic nature of B-F bonds in BF_3 is

- A) large electronegativity of F B) three centred two electron bonds in BF_3
 C) $pp - dp$ back bonding D) $pp - pp$ back bonding
15. The product formed in the reaction, $BCl_3 + H_2O \rightarrow$ product is :
 A) $H_3BO_3 + HCl$ B) $B_2O_3 + HOCl$ C) $B_2H_6 + HCl$ D) no reaction
16. Reactivity of borazole is greater than that of benzene because :
 A) borazole is non-polar compound B) borazole is polar compound
 C) borazole is electron deficient compound D) of localized electrons in it
17. In the electrolysis of alumina, cryolite is added to:
 A) lower the melting point of alumina B) increase the electrical conductivity
 C) both (a) and (b) D) remove impurities from alumina
18. Mineral of Al that doesn't contain oxygen
 A) corundum B) diaspor C) bauxite D) cryolite
19. Aluminum metal is corroded in coastal places near to the sea, because protective oxide film:
 A) is removed by sea water B) reacts with sea water
 C) is attacked by salts presents in sea water D) reacts with sand parties
20. Duralumin is an alloy of :
 A) Al and Mg B) Mg and Cu C) Al, Mg, Mn and Cu D) Al and Cu
21. When Al is added to potassium hydroxide solution :
 A) no reaction takes place B) oxygen is evolved
 C) water is produced D) hydrogen is evolved
22. Aqueous ammonia is used as a precipitating reagent for Al^{3+} ions as $Al(OH)_3$ rather than aqueous $NaOH$, because:
 A) NH_4^+ is a weak base B) $NaOH$ is a very strong base
 C) $NaOH$ forms $[Al(OH)_4]^-$ ions D) $NaOH$ forms $[Al(OH)_4]^+$ ions
23. The dissolution of $Al(OH)_3$ by a solution of $NaOH$ results in the formation of :
 A) $[Al(H_2O)_4(OH)]^{2+}$ B) $[Al(H_2O)_2(OH)_4]^-$ C) $[Al(H_2O)_3(OH)_3]$ D) $[Al(H_2O)_6(OH)_3]$
24. Which reaction can not give anhydrous $AlCl_3$?
 A) Heating of $AlCl_3 \cdot 6H_2O$ B) Passing dry HCl over heated aluminum powder
 C) passing dry Cl_2 over heated aluminum powder
 D) Heating a mixture of alumina and coke in a current of dry Cl_2
25. Which of the following is pseudo alum ?
 A) $(NH_4)_2 SO_4 \cdot Fe_2(SO_4)_3 \cdot 24H_2O$ B) $K_2SO_4 Al_2(SO_4)_3 \cdot 24H_2O$
 C) $MnSO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ D) None of these
26. Thermite, a mixture used for welding is :
 A) Fe and Al B) ferric oxide and aluminium powder
 C) barium peroxide and magnesium powder D) Cu and aluminium
27. Aluminium vessels should not be washed with materials containing washing soda because
 A) washing soda is expensive B) washing soda is easily decomposed
 C) washing soda reacts with aluminum to form soluble aluminate
 D) washing soda reacts with aluminium to form insoluble aluminium oxide
28. The structure of diborane contains :
 A) four 2C-2e bonds and two 3C-2e bonds B) four 2C-2e bonds and two 3C-2e bonds

- C) two 2C-2e bonds and two 3C-2e bonds D) four 2C-2e bonds and two 3C-2e bonds
29. Which one of the following is the correct statement ?
 A) Boric acid is protonic acid
 B) Beryllium exhibits coordination number of six
 C) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase
 D) $B_2H_6 \cdot 2NH_3$ is known as 'inorganic benzene'
30. A substance which gives a brick red flame and breaks down on heating giving oxygen and brown gas is :
 A) calcium carbonate B) magnesium nitrate
 C) magnesium carbonate D) calcium nitrate
31. Mortar is a mixture of :
 A) $CaCO_3$ and CaO B) slaked lime and water
 C) slaked lime, sand water D) Sand & CaO
32. The substance not likely to contain $CaCO_3$ is :
 A) sea shells B) dolomite C) a marble statue D) calcined gypsum
33. Which compound gives acetylene on reaction with water :
 A) Al_4C_3 B) Mg_3N_2 C) CaC_2 D) CaH_2
34. Chemical A is used for water softening to remove temporary hardness. A reacts with sodium carbonate to generate caustic soda. When CO_2 is bubbled through A, it turns cloudy. What is the chemical formula of A?
 A) $CaCO_3$ B) CaO C) $Ca(OH)_2$ D) $Ca(HCO_3)_2$
35. Ca on exposure in moist air forms a layer on surface of :
 A) $CaCO_3$ B) $Ca(OH)_2$ C) $CaCO_3 \cdot Ca(OH)_2$ D) CaO
36. Substance which absorbs CO_2 and violently reacts with H_2O with sound is :
 A) $CaCO_3$ B) CaO C) H_2SO_4 D) ZnO
37. The compound X on heating gives a colourless gas. The residue is dissolved in water to obtain Y. Excess CO_2 is bubbled through aqueous solution of Y, Z is formed. Z on gentle heating gives back X. The compound X is
 A) $CaCO_3$ B) Na_2CO_3 C) $CaSO_4 \cdot 2H_2O$ D) K_2CO_3
38. Setting of plaster of paris involves :
 A) Oxidation with atmospheric oxygen B) combination with atmospheric CO_2
 C) dehydration D) hydration to yield another hydrate
39. At high temperature, nitrogen combines with CaC_2 to give :
 A) calcium cyanide B) calcium cyanamide
 C) calcium carbonate D) calcium nitride
40. The ionic carbide is:
 A) CaC_2 B) ZnC C) SiC D) TiC

REASONING TYPE

41. **Statement – 1:** Anhydrous $AlCl_3$ is covalent while aqueous $AlCl_3$ is ionic.
Statement – 2: Due to greater hydration energy of Al^{3+} ion in water.
42. **Statement – 1:** Concentrated HNO_3 renders the 'Al' metal passive.
Statement – 2: It is an oxidizing agent and produce a protective layer of oxide on the surface of 'Al' metal.

